

Recovery

Merge Readiness Score

32.5 out of 100

NOT MERGEABLE

Gate: blocked

Not mergeable. Measured statement coverage is 16.7%, under 20%, and the Merge Readiness Score is 32.5, under 40.

Score factors

Factor	Earned	Max	Detail
Structural coverage	5.69	40	Statement coverage 16.7% and branch coverage 10.5% combine to 14.2 before the 40-point weight. Tool percent_covered is 14.1% and is not used: branch mode blends arcs into that headline.
Property-based resilience	25	25	Ran 1000 property trials across structural conformance, determinism, and entity substitution. Failing inputs were shrunk toward simpler values.
Prompt / semantic alignment	0.26	15	ΔH coverage 1.7%. ΔH is the drop in Shannon entropy of specification tokens after removing tokens named by tests. This is a lexical spotlight, not transformer attention.
Blast radius safety	1.57	10	No diff was supplied. The radius is the whole indexed graph, so union coverage collapses to coverage of those functions.
Execution-time efficiency	0	10	Mann-Whitney U compared baseline inputs with heavier inputs on this same revision. A regression is recorded only when $p < 0.05$ and the median is more than 8x slower. This is not a cross-commit EffiBench run. Heavier inputs were slower for subtotal (10.99x).

Coverage by module

Vector bars drawn by Typst from the coverage JSON. Not a placeholder.

shop.cart		45%
shop.payments		0%
shop.pricing		0%

Gap severity

critical		2
high		1
medium		2
low		0

Property-based testing

Ran 1000 property trials across structural conformance, determinism, and entity substitution. Failing inputs were shrunk toward simpler values.

Symbol	Trials	Passed	Failed
subtotal	250	250	0
apply_coupon	250	250	0
charge	250	250	0
refund	250	250	0

Mutation, prompt coverage, blast radius, timing

Killed 1/5 viable mutants on a temp copy (0 timeouts counted as killed, 0 unviable excluded). MSI 20.0. MSI_total 20.0 keeps unviable mutants in the denominator. This is a sampled operator flip, not a mutmut or cosmic-ray campaign. Project files were not edited.

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Authenticity scorecard

Authenticity scorecard is not the Merge Readiness Score and does not move the gate. DAR checks the workspace, the stdlib, and declared manifests. It does not query a package registry. FIRI is a static scan, not a reproduction of a flake.

Dimension	Value	Threshold	Status
Dependency authenticity (DAR)	100.0	100%	PASS
Statement coverage	16.7	$\geq 80\%$	FAIL
Branch coverage	10.5	$\geq 75\%$	FAIL
Mutation score (MSI)	20.0	$\geq 70\%$	FAIL
Assertion strength (ASR)	100.0	$\geq 85\%$	PASS

Mean CRAP	34.2	≤ 15	FAIL
CRAP > 30	3	0	FAIL
Flakiness risk (FIRI)	0.0	0%	PASS

Autonomous suggestions

- subtotal has PageRank 0.2000 in community 0. Test that symbol against its graph neighbors instead of an isolated file dump.
- charge has CRAP 56.0 (complexity 7). Above 30, add tests or split the function.
- charge: Add a pytest test that calls charge(amount, method, authorized) and asserts each branch (6 static decision points).
- refund: Add a pytest test that calls refund(amount, captured) and asserts each branch (2 static decision points).
- tier_price: Add a pytest test that calls tier_price(units, tier) and asserts each branch (6 static decision points).

Rubric

Recoverage score rubric

The score is the Merge Readiness Score (MRS), 0–100. It is the PRD's weighted sum, not a line-coverage percentage.

Factor	Points	How it is measured
Structural coverage	40	60% project statement coverage + 40% branch coverage. Unmeasured runtime coverage scores 0. coverage.py's own `percent\_covered` is reported beside it and is not the score: with `--branch` that headline blends arcs into statements.
Property-based resilience	25	Share of in-process property trials that held. The properties are structural conformance, determinism, and entity substitution. The engine runs at least 1,000 inputs when pure functions exist. Failing inputs are shrunk. This is not a frozen expected-output check.
Prompt / semantic alignment	15	ΔH , the drop in Shannon entropy of specification tokens once tokens named by tests are removed. Offline this is a lexical spotlight, not transformer attention. No spec text scores 0.
Blast radius safety	10	Union coverage of the call-graph radius. With no diff, the radius is the whole indexed graph and the report says so.
Execution-time efficiency	10	Mann-Whitney U between baseline inputs and heavier inputs on this revision. Points are lost only when $p < 0.05$ and the median is more than 8× slower. This is not a cross-commit benchmark.

Mutation testing is a separate measurement. When it runs, mutants are applied on a temp copy and the kill count is printed. It is not silently folded into MRS. If it does not run, the report says it did not run.

Flake markers found by a static scan subtract 2 points, floored at 0. That scan is not a reproduction. Drafts that `recoverage generate` keeps are executed 5 times in a sandbox before they are copied back.

Gates

MRS is mapped onto four gates so a CI job can fail below a named bar. The PRD badge uses the same threshold, default 85.

Gate	Rule
blocked	No test runner, or measured project statement coverage is below 20%, or MRS is below 40. Badge: NOT MERGEABLE.
needs-review	A runner exists and MRS is at least 40, but the merge bar is not met. Unmeasured runtime coverage cannot be graded above this gate.
merge-ready	$MRS \geq 70$, statement coverage $\geq 60\%$, runtime coverage was measured, and there is no critical gap. Badge: MERGEABLE.
production-ready	$MRS \geq 85$ (the PRD threshold), statement coverage $\geq 60\%$, runtime coverage was measured, and there is no critical gap. Badge: PRODUCTION READY.

`--threshold` accepts a gate name or a number. A gate threshold passes only when the result gate ranks at least that high: blocked < needs-review < merge-ready < production-ready. A numeric threshold passes when MRS is greater than or equal to the number.

Separate from MRS

CRAP, assertion strength, dependency authenticity, static flakiness risk, and the mutation score are reported on an authenticity scorecard. They are not added into the 100 points and they do not move the gate. A generated draft is kept when it adds covered lines or kills a mutant the existing suite left alive. That filter is there because a coverage bump with a vacuous assertion is not evidence. Drafts still do not lock in an observed return value.